

Wood-Cased Pencils Vs. Mechanical



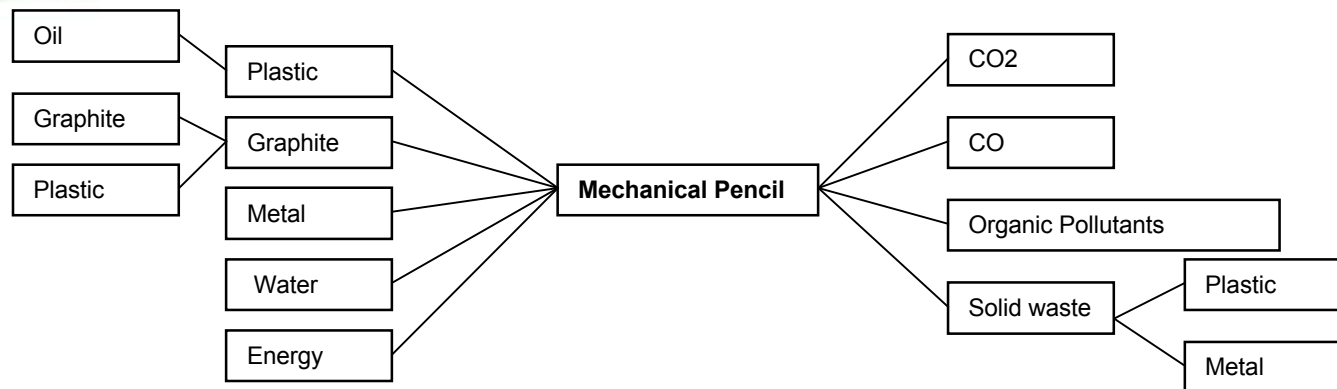
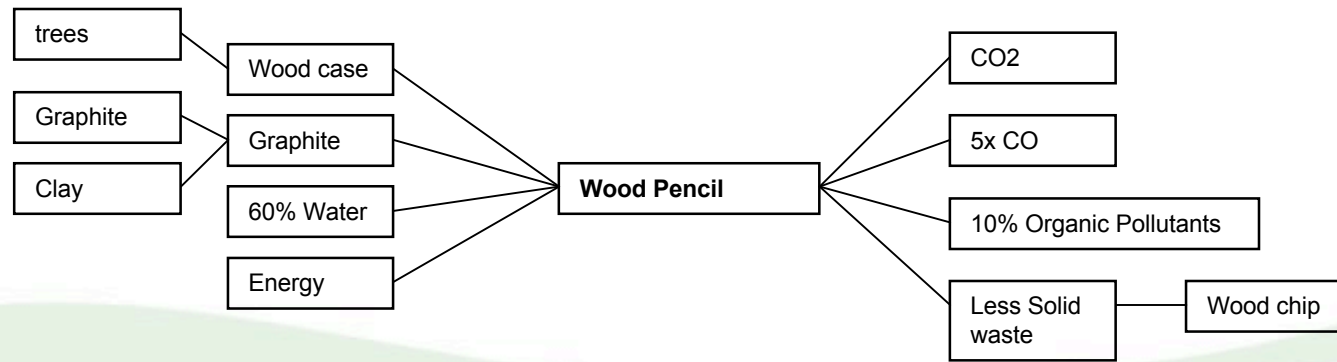
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Renewable Vs Reusable

- Wood is a renewable resource, plastic is not
- Wooden pencils have a limited lifespan
- Plastic pencils can be refilled and reused for years
 - Life limited only by misplacement and destruction
- Writing instrument manufacturers association
 - Over 3 billion wooden
 - 550 million mechanical
- What's better overall?
 - Renewable materials or reusable product?



Life Cycle Analysis

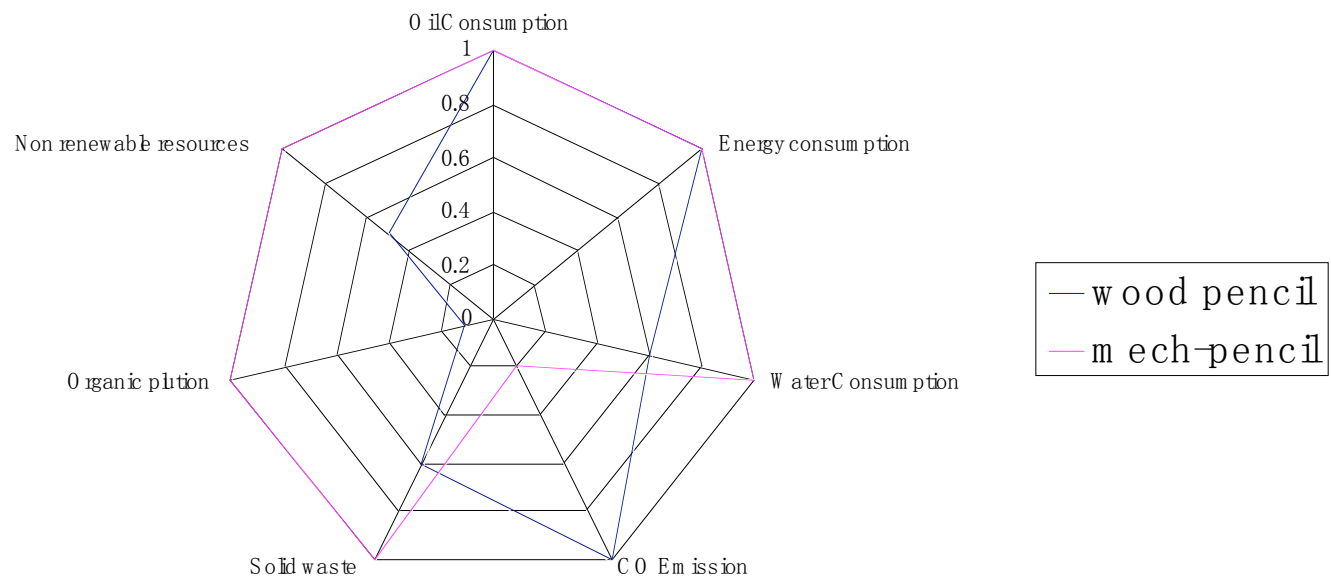


Life Cycle Analysis

- Wood cased pencils
 - 4 time more raw material consumption
 - Similar energy consumption
 - 5 to 6 times more CO emissions
- Plastic pencils
 - Twice the non-renewable resource materials
 - 40% more water consumption
 - More non-renewable energy used
 - 90% more organic pollutants emitted
 - Greater waste water effluents
 - More net process solid waste
 - Significantly more hazardous waste



Impact Comparison



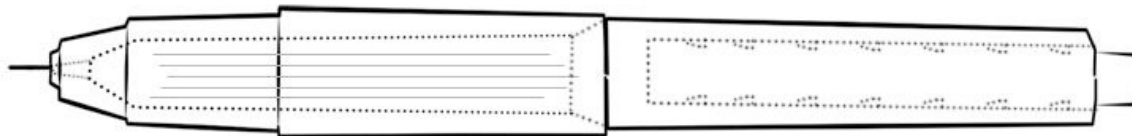
Which is Better Overall?

- Individually less environmental impact to make wooden pencils than mechanical
- Plastic pencils can last much longer than wooden pencils
- Environmental impacts similar enough for life-span to make a difference
- When refilled as intended, plastic pencils have a smaller impact



A Better Solution

- Extend the life of plastic pencils even more
 - Larger eraser
 - More graphite included in barrel
 - Higher quality
 - Discourage misplacement
- Reduce wastes
 - Minimize packaging



Questions?

